

西南财经大学本科期末考试试题册（B）

（2021—2022学年第1学期）

以下各项由命题教师填写：

课程名称：面向对象程序设计（JAVA SE）(英) 命题教师：王磊

适用对象（年级专业）：2021 信息管理与信息系统专业

使用试题的任课教师姓名：王磊

试题说明：

- 1、考试类型：闭卷[] 开卷[√]
- 2、本套试题共 两 道大题，共 6 页，完卷时间 120 分钟。
- 3、考试用品中除纸、笔、尺子外，可另带的用具有：
计算器[] 字典[] 教材1本 [√]等
(请在下划线上填上具体数字或内容，所选[]内打钩)

考试时间（由制卷方填写）：

以下各项由学生填写：

任课教师：

年级专业：

学生姓名：

学 号：

- 考生注意事项：
1. 出示学生证（或身份证）和准考证于桌面左上角，以备查验。
 2. 拿到试卷后清点并检查试卷页数，如有重页、页数不足、空白页及印刷模糊等举手向监考教师示意调换试卷。
 3. 答题前先将试题册及答题纸上的任课教师、专业、年级、学号、姓名填写完整。
 4. 所有答案均需填写在答题纸上，答在试题册上无效。
 5. 考生不得携带任何通讯工具进入考场。
 6. 考试结束后，将试题册、答题纸和草稿纸等下发材料上交监考教师，不得带离考场。
 7. 严格遵守考场纪律。

Part I: Single Choice Questions:

Mark your answers on below sheet (total 30 pts, 3 pts each)

Answer sheet:

1	2	3	4	5	6	7	8	9	10

1. Which of the following data types **cannot** be used as the condition for a **switch** statement?

- A. char
- B. boolean
- C. byte
- D. int

2. Which of the following expressions is equivalent to the **boolean** expression?

!(A < 5 && B != C)

- A. A > 5 || B != C
- B. A >= 5 && B == C
- C. !(A < 5) || (B != C)
- D. A >= 5 || B == C

3. Answer the next 2 questions, considering the following code fragment:

```
int sum = 0;
int i = 0;
while (i < 5)
{
    sum = sum + i;
    i++;
}
System.out.print(i);
System.out.print(" ");
System.out.print(sum);
```

What is the value of i when System.out.print(i) is executed?

- A. 6
- B. 5
- C. 4
- D. 3

4. What is the value of sum when System.out.print(sum) is executed?

- A. 6
- B. 21
- C. 10
- D. 15

5. What is the final value of x when the following code fragment is executed?

```
int x = 0;
for (int i = 0; i < 10; i++) {
    if (i % 4 == 0)
        x += 10;
    else if (i % 5 == 1)
        x += 30;

    if (i % 3 == 0)
        x += 50;
}
```

- A. 250
- B. 280
- C. 290
- D. 310

6. How many elements are in the array y when the following code fragment is executed?

```
long [] x = new long[10];
long [] y = {1,2,3,4,5};
x=y;
```

- A. 5
- B. 6
- C. 9
- D. 10

7 Which of the following statements correctly creates an array of 5 empty Strings?

(1). String a[] = new String[5];

```
for (int i=0; i<; i++)
```

```
    a[i]=" ";
```

(2). String a[]={ " ", " ", " ", " ", " "};

(3). String a[5];

(4). String [5] a;

(5). String [] a= new String[5];

```
    a={" ", " ", " ", " ", " "};
```

A. None

B. Only (1)

C. Only (1) and (2)

D. Only (1) (2) and (5)

8. What type of relationship exists between someMeth in classes A and someMeth in class B?

```
class A{
```

```
    private void someMeth(){
```

```
        System.out.println( "from class A" );
```

```
    }
```

```
}
```

```
class B extends A{
```

```
    public void someMeth( String x ){
```

```
        System.out.println( "from class B: " + x );
```

```
    }
```

```
}
```

A. method overriding

B. method overloading

C. both method overriding and method overloading

D. neither method overriding nor method overloading

9. Suppose the class **Undergraduate** extends the class **Student** which extends the class **Person**.

Given the following variable declaration:

```
Person p = new Person();
```

Student s = new Student();

Undergraduate ug = new Undergraduate();

Which of the following assignments are legal?

I. p = ug;

II. p = new Undergraduate();

III. ug = new Student();

IV. ug = p;

V. s = new Person();

A. III and IV

B. I and IV

C. I and II

D. I, II and V

10. Which of the following is true about an abstract method inherited into a class XY?

A. It must be defined in XY before XY can be instantiated.

B. It always forces XY to become abstract.

C. It overrides any method in XY with the same name.

D. None of these is true.

Part II: Write Programs.(total 70 pts)

1. (10pts) Write a program that prints numbers from 1 to 20, but if numbers are multiples/times of 5, print character F instead, and if numbers are multiples/times of 6, print character S instead.

2. (10pts) Write a program that prompts the user to enter an integer n (assume $n \geq 1$) and displays a pattern on the screen as follows (suppose that $n=5$).

```
1
1 4
1 4 9
1 4 9 16
1 4 9 16 25
```

3.(15pts) Write a method to invert the order of a array using the following header:

```
public static void invertArray(int [ ] x){  
    }  
}
```

In the main method, use the invertArray method to invert the order of a given array b (e.g., int b[] = {1,2,3,4,5}) and display the content of it (e.g.: 5 4 3 2 1).

4.(15pts) Design a class named **GeometryObject** , and its two data members centerX, centerY denotes its central position in the coordinate system. Make **Rectangle** subclass of **GeometryObject**. A Rectangle object has data members of length and width, representing the length and the width of it, and it also has the below method members:

- At least one constructor
- Method area() to calculate the area of a Rectangle object.
- Method crossArea(Rectangle a) to return the overlapping area with Rectangle object a if they overlap each other, otherwise outputs 0.

In the main method, create a Rectangle object *rectA*, output its area and the overlapping area with another Rectangle object *rectB*.

5. (20 pts) Write a GUI program to display BMI and related interpretation when a person input his weight (in pounds) and height (in inches) and press down the button. The program may run like the below figure.



The screenshot shows a Java FX GUI window titled "BMI Calculator". It features standard window controls (minimize, maximize, close) in the top right corner. The main area contains two input fields: "Weight(pounds):" with the value "146" and "Height(inches):" with the value "70". To the right of these fields is a button labeled "Compute BMI". Below the input fields, the results are displayed: "BMI : 20.948494" and "Interpretation: Normal!".

Requirements:

- Use suitable Java FX GUI components.
- Use suitable layout to organize components.
- At least contains one event handling process.